

Automated Crowd Management in Bus Transport Service

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Abstract—Crowd Management in Bus Transport Service provides an easy way to know about the live crowd density present inside the bus to the commuters present at the upcoming bus stop. This paper presents an Automated Crowd Management System using the algorithms of Machine Learning and IoT Technologies. The crowd density is detected inside the bus and is classified in to 5 different levels which will eventually be displayed on the LCD screen at the bus stop. The proposed system has an accuracy of 93.09% and the time required to transmit the crowd density to the bus stop is just a few seconds. This thus, provides commuters with a safe journey and helps them know about the live crowd density in a particular bus and help them plan accordingly as to which bus to board or look for an alternative mode of transport.

Keywords—crowd, CNN, Raspberry Pi, LCD, Bus, density, server

I. INTRODUCTION

In case of public transports, it is usually noticed that some of the buses are overcrowded and at the same time some of them are half-empty although they are heading towards the same destination at the same time.

The problem gets even worse if the bus is late. People try to board the bus that comes first as they have no idea about the upcoming buses. The buses coming thereafter aren't fully occupied. This uneven crowd density in the buses is undesirable.

Most of the times, this uneven crowd density has caused numerous human deaths along with many casualties. There are also situations where people have had narrow death experiences due to overcrowding.

There is a need to develop a system that detects the density inside the bus and displays the crowd density level at the bus stop. Knowing about the crowd density in the upcoming buses, commuters will be in a position to decide whether to board the bus or wait for the next bus. This will help the corporations administering the system to manage the crowd.

Currently, in the present scenario, there is no such existing system. Some of the android applications such as Google maps follow a procedure of obtaining feedback from the passengers after they have travelled on a particular bus. They tend to ask the passengers about the level of crowd density present in the bus. This information obtained is highly unreliable. There can be a situation wherein the information

that is fed is incorrect or people might not even provide feedback due to their busy schedules. Our system does not depend on the reviews given by passengers. The installation of a camera along with the dedicated system inside the bus prevents any human interaction with the system thus providing an accurate and reliable output. The camera independently captures the image of the crowd and processes it to the system which detects the crowd level and updates it onto the server. The server then automatically updates the crowd density level at the LCD screen present at the bus stop.

The organization of the paper is as follows: Section 2 provides a summary of the related work. Section 3 states the proposed solution for managing the crowd on the bus. Section 4 elaborates the solution. Section 5 discusses the results obtained from the working solution. Section 6 concludes the paper along with the future scope.

II. RELATED WORK

The current scenario of public transport is not viable for commuters, especially during peak hours. Most of them face the problem of overcrowding and have little to no idea about the crowd density inside the bus as well as the real-time location of the bus.

In [1], the authors propose a design which consists of micro-controller, IR proximity sensor, GPS Module, IoT Module. The system uses IR proximity sensors to detect the crowd present inside the bus and use GPS to get the live location of the bus which are displayed on a webpage. The major drawback of this system is that it has employed IR sensors at the front and back of the doors. This can give a wrong count of people present inside the bus since there can be a case that a person enters and exits the bus from the front or back door itself.

The authors of [2] propose a system that counts the number of heads in an area. The system uses Raspberry Pi for human tracking and implements Haar Cascade Amplifier to calculate the number of heads. This system is highly disadvantageous in situations of high crowds because the time required to calculate the number of people in public transport will increase, thus making it a cumbersome task.

In [3], authors present a system that uses the RANSAC (Random Sample Consensus) algorithm which is used to estimate the crowd flow direction and classify people into different groups. The "Lucas-Kanade (LK) algorithm" is implemented to determine the optical flow of a crowd. The system classifies people based on how they are located in that area. The crowd present at the centre are classified as inliers whereas the crowd present on the edges or boundaries are known as outliers. But this method is limited to only short distances and detects the heads that are present in a particular area. Also, people who are scattered around the place are not taken into consideration.

Ref. [4] presents a multiscale convolutional neural network (MSCNN) for single image crowd counting instead of the conventional multi-column or multi-network model to extract the scale relevant features. These conventional methods have increased computation and complicated optimization. The use of multi-scale blobs in a single column network had produced a much accurate and robust performance.

In [5], the authors have developed a system that includes Multi Convolutional Neural Networks (MCNN) instead of the conventional Convolutional Neural Networks (CNN). This helps to detect the sum of people accurately. But this system does not validate the cross-scene data. This creates uncertainty while deploying the system in real-life situations. In [6] and [7], the authors present an overview of the various ways in which a crowd can be detected. They have divided the approaches for crowd estimation into three main categories: detection based, regression-based and density-based. Apart from that, they have included various CNN based estimation techniques.

In [9], the authors have proposed a technique for public safety in large gatherings which uses colour-based segmentation and generic head detector. This technique is majorly used for video surveillance. In [10], the authors have presented a review on people counting and crowd estimation using computer vision. Their method has direct and indirect based approaches. The direct based approach counts the number of people simultaneously as they are segmented correctly whereas the indirect approach includes developing learning algorithms which relate to certain features that help in crowd estimation and counting. In [11], the authors have developed a system wherein they can estimate high crowds in still images. They have used a combination of methods such as interest points (SIFT), Fourier analysis, wavelet decomposition, GLCM features and low confidence head detections, to estimate the counts. In [12] the authors have devised a technique that is based on images. They have used filters to model the density maps for various sizes of heads. The images that are used had random sizes.

In [13], the authors have developed a procedure that counts the number of people using an adaptive background model. In [14], the authors have devised a technique which is used to count and locate the individuals in a low-resolution

image. They have implemented a method that consisted of three steps. The first step is to perform postprocessing on the background to roughly estimate the number of people. Second, an expectation minimization technique which is developed to indicate the number of individuals in low resolution and the third step is to locate individuals using feature points. In [15], the authors have detected the passenger flow. They have used Haar feature and Adaboost algorithm to detect the heads of the passengers. This method is generally used for human tracking which might violate a person's privacy which is not happening in our model.

Most of the papers discussed above have used CNN and IR sensors to detect the number of people. The objective of these papers was mainly centred on counting the number of heads present in a particular frame. This is a complex process in case of highly dense crowds the number of heads cannot be counted individually and it becomes a time-consuming process. Hence, our proposed system will calculate the crowd density that is present inside the bus and will display the estimated crowd density inside the bus at the bus stop.

III. PROPOSED SYSTEM

The proposed system is to detect the live crowd inside the bus and display the real-time density of the crowd at the bus stop. Every bus will have a camera which will capture the image of the crowd density of the bus and the image will be processed on the bus. The density of crowd in the bus will be classified in one the class out of five viz. Very Low, Low, Moderate, High, Very High. The detected crowd level will be transmitted to the server. This result will be displayed on the LCD screens installed on the bus stops. The crowd density will be updated after every predetermined time interval depending on the distance between two stops.

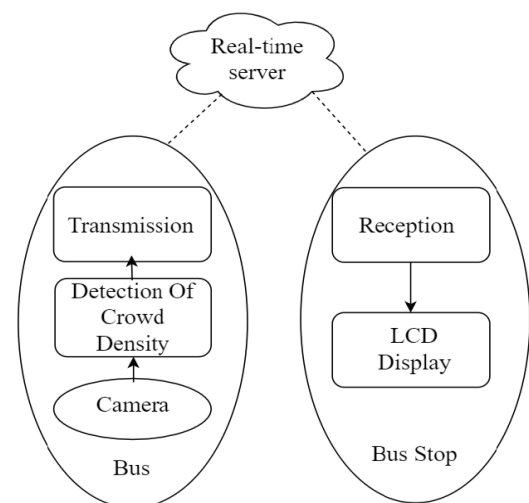


Fig.1. Proposed Design

IV. METHODOLOGY

A. Dataset-

To classify the crowd density inside the bus, Convolutional Neural Networks (CNN) have been used. The dataset to train our model was collected while travelling inside the bus. A camera is installed at the top right corner of the bus. An entire video of different crowd levels was collected. This video was converted into frames. To classify the crowd density into five different levels, the frames are tagged accordingly. The dataset has 6000 images in total. 4500 images have been used as training dataset and 1500 images for the testing dataset. The dataset is based on real-life situations and is hence practically implementable.

B. CNN Model-

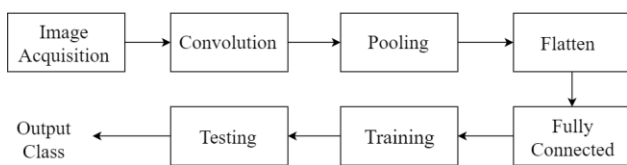


Fig. 2. CNN Model

1) Image Acquisition-

The very first step in our proposed system is to capture the image of the live crowd density present inside the bus. To capture the image of the crowd, a camera is installed at the back of the bus on the top corner. This will capture the maximum area of the bus thus capturing the maximum amount of crowd. The image of the crowd is then sent to the CNN model which will detect the crowd density level.

2) Convolution-

The first layer of a CNN model is the convolution layer. This layer is responsible for extracting the high level features from the images. The first convolution layer usually extracts the low level features such as gradient, colors, edges etc. As the number of layers is increased, the features that are extracted are of high complexity. This helps the model to visualize the dataset. ReLu(Rectified Linear unit) activation is added to increase the non linearity in the images[8].

3) Pooling-

This layer is essentially responsible for reducing the computational power that is required to process the data. For our model, the Max Pooling layer is used which extracts the dominant features and acts as noise suppressant.

4) Flatten-

This layer is used to convert all the data into a single one-dimensional layer. This helps to easily input the data to the fully connected layer.

5) Fully Connected Layer-

The dense layer is used to add the fully connected layer to the neural layer. The fully connected layer takes input from the previous layer and correlates it with the most probable layer by taking into account the product between the weights and previous layer.

6) Training-

Training involves making our dataset fit the parameters of our model. It involves backpropagation, learning, loss function and forward path.

7) Testing-

Testing is used to assess the performance of our model. This is essentially helpful in knowing whether the test image corresponds to the ground truth.

C. Implementation-

Our model consists of installing a webcam on the top corners of the bus along with the dedicated system. The camera is interfaced with the Raspberry Pi. The camera is programmed so that it captures the image at a predefined time interval. The image captured by the camera is fed to our dedicated system. The CNN model processes the image and predicts the crowd density level. The defined output is updated on our real-time server. This real-time server transmits the acquired output to the system present at the bus stop. This system consists of a Raspberry Pi interfaced with the LCD. As soon as the system receives the data it displays it on the bus stop. This crowd density level is updated at regular time intervals so that the commuters get an idea about the crowd density present in the bus at any particular time.

D. Components-

1) Raspberry Pi-

In this project, different versions of Raspberry Pi (Rpi) is included. The one at the transmitter side is the latest version of Raspberry pi i.e. Rpi 4 Model B and the one at receiver side is Rpi3 model B+.

The Rpi 4 is used as a dedicated system inside the bus which is interfaced with the camera. As soon as an image is captured by the camera, it is processed inside the Rpi 4 which transmits the crowd density to the server. The Rpi 3 is interfaced with the LCD screen at the bus stop. This Rpi receives the transmitted crowd density result from the server and displays it on the LCD.



Fig.3. Raspberry Pi

2) LCD Display-

The 2X16 LCD screen is used. The LCD is interfaced with the Rpi 3 and displays the result. The first row of the LCDs displays the license plate of the bus along with its bus number. The second row displays the real-time crowd density inside that particular bus.



Fig.4. 2X16 LCD

3) Webcam-

Webcam is used to capture the crowd images in the bus and is interfaced with Rpi. This camera is programmed for a specified time interval. After this specified time the camera will capture the image of the live crowd present inside the bus. This camera is interfaced with the Rpi 4, hence as soon as it captures the image, the image is then given as input to our trained model.



Fig.5.Webcam

4) Firebase-

Google Firebase is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. A real-time database is required in this project. Firebase provides offline support as well. This is because of Firebase SDKs data to our disk. Once connectivity is re-established, the client device receives any changes it has missed and synchronizes it with the current server state.

For our project, Google Firebase has been used as the main server. Both of our Rpi is connected with the real-time server. When the dedicated system inside the bus updates the result onto the server, automatically the receiver's side gets notified about it. Without much delay, the receiver updates the changed crowd density. Firebase has helped us overcome the problem of distance between the bus and the passengers. As it is a real-time server, hence there are no connectivity issues between the bus and the bus stop thus, helping us transmit our information effortlessly.

V. RESULTS

The developed system works as expected and interacts with all the components properly. Our system has been tested in real-life situations and it predicts the output accurately. The system is now capable of predicting one of the density levels amongst five with an accuracy of approximately 90%. Below are some of the sample test images for different density levels.



Fig.6. a. Very Low-Density



Fig.6. b. Very High Density



Fig 6.c. Moderate Density



Fig 6.d. High Density

The images below are depicting the result obtained at the receivers 'end,



Fig.7. a. Very Low-Density



Fig.7.b-Very High Density



Fig.7. c. Moderate Density



Fig.7. d. High Density

The confusion matrix for the training images has been shown in Table 1. The results for every class are around 90% which proves the credibility of our model. The training accuracy for our model is greater than our testing accuracy. From the confusion matrix, the model gives a precise output which can be easily deducted. Increasing the dataset will give us even more precise output.

TABLE I. CONFUSION MATRIX FOR TRAINING IMAGES

Desired/ Output	Very High	High	Moderate	Low	Very Low
Very High	833	69	0	0	0
High	60	826	66	0	0
Moderate	7	5	830	67	6
Low	0	0	4	828	71
Very Low	0	0	0	5	823
Percent Correct	92.55	91.77	92.22	92	91.44

The confusion matrix for all the testing set images has been shown in Table 2. The confusion matrix essentially gives the accuracy of each class along with the number of false positives and false negatives. For our system, all the classes have an average accuracy of 90.53%. This itself proves that our system has an accuracy of about 90% thus, making our system more reliable.

TABLE II. CONFUSION MATRIX FOR TESTING IMAGES

Desired/ Output	Very High	High	Moderate	Low	Very Low
Very High	270	7	2	0	0
High	26	265	14	3	5
Moderate	2	25	277	14	4
Low	2	3	7	280	25
Very Low	0	0	0	3	266
Percent Correct	90	88.33	92.33	93.33	88.66

The real-time results along with the confusion matrix validate that our system is quite accurate and precise in detecting the crowd density. The detected crowd density is easily transmitted to the server with very less delay thus, preventing any time lag.

VI CONCLUSION AND FUTURE SCOPE

The system helps let the commuters know about the live crowd density of a bus. It accurately identifies the crowd density levels from the captured images.

Such a system will cater to the needs of people who use public transport for their day-to-day needs. As a real-time server have been used, hence there will be no connectivity issues in transmitting the data about crowd density. Also, since images are just getting captured and not recognizing a person's face hence there would not be an issue about privacy invasion.

The proposed system can further be improved by incorporating GPS in it so that people will get an idea about how much time the bus will take to reach the required destination. An application can also be developed so that people will know the crowd density of a bus while they are at their workplace or while they are at their homes.

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